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(54) **HEAT SINK MANUFACTURING METHOD**

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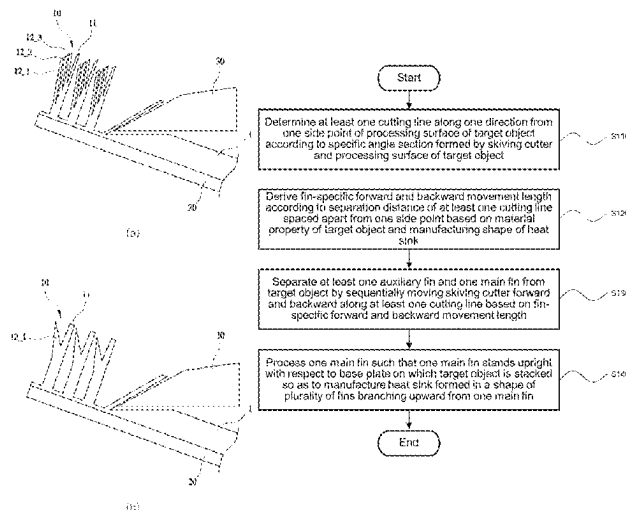
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B26D 3/122; F28F 3/048

See application file for complete search history.

(57) **ABSTRACT**

Disclosed is a heat sink manufacturing method including the steps of: determining at least one cutting line in one direction from one side point on the processing surface; individually deriving fin-specific forward and backward lengths of the skiving cutter according to the separation distance of the at least one cutting line separated from the one side point; separating, on the basis of the fin-specific forward and backward lengths, at least one auxiliary fin and one main fin from the object by sequentially moving the skiving cutter forward and backward along the at least one cutting line; and processing the one main fin so that same stands upright relative to a base plate on which the object is stacked, so as to manufacture a heat sink formed in the shape of a plurality of fins branching upward from the one main fin.

4 Claims, 7 Drawing Sheets



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FIG. 1

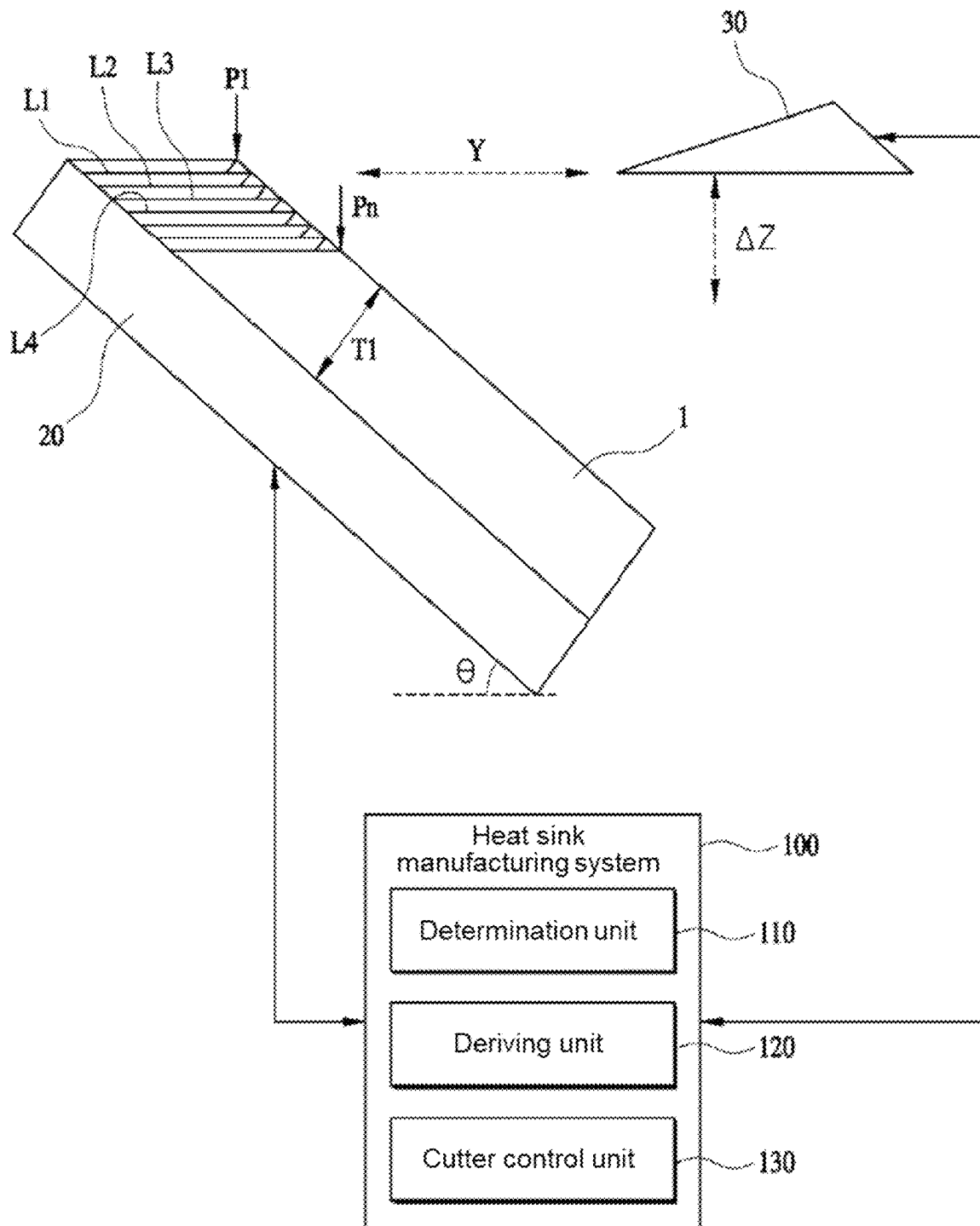
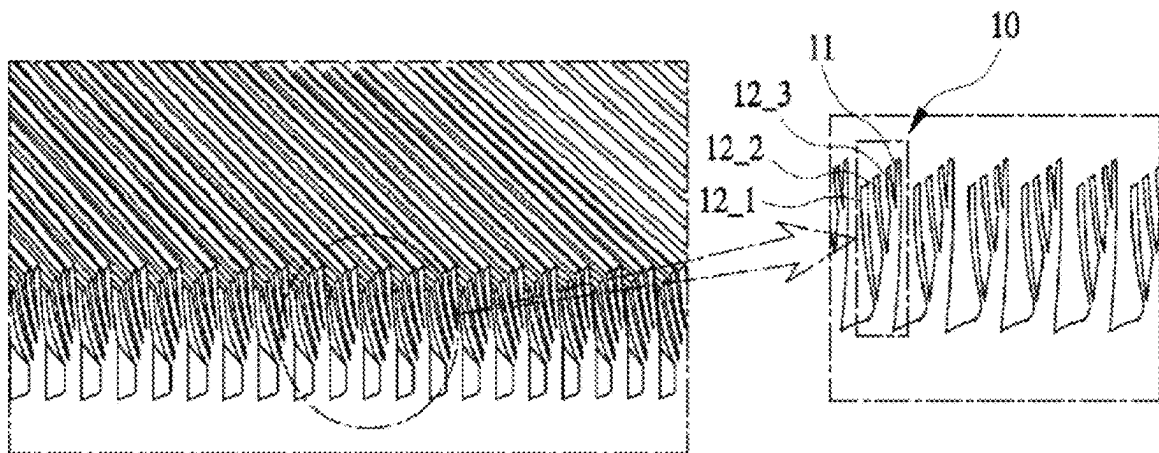
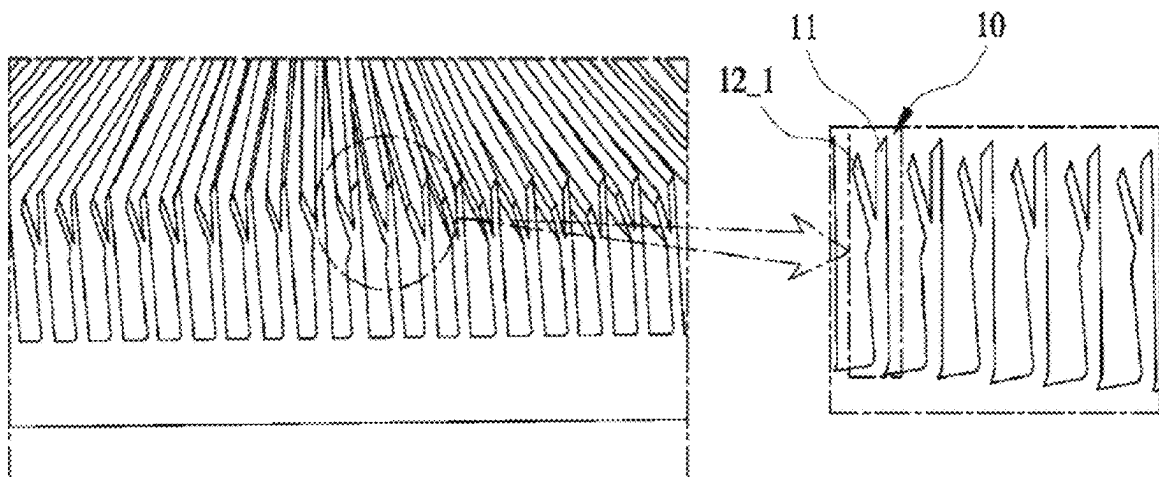


FIG. 2

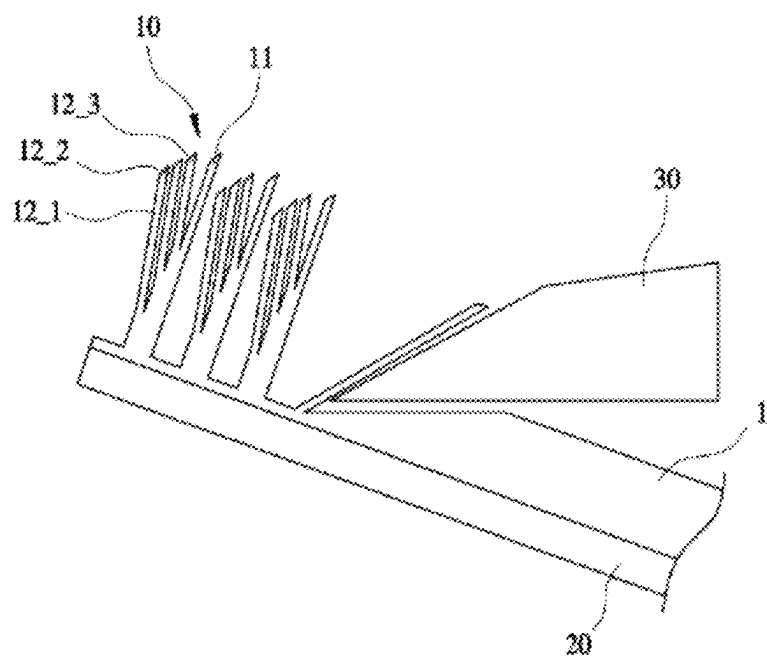


(a)

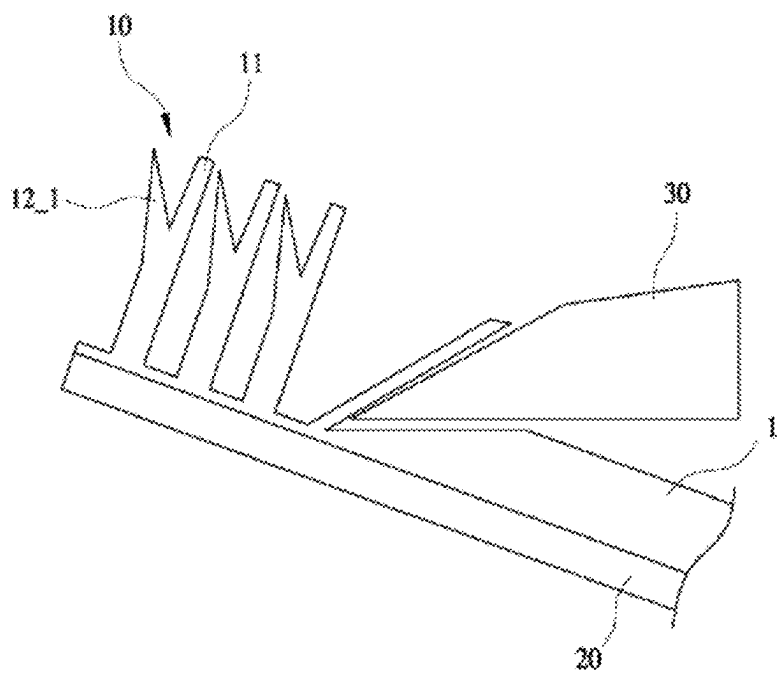


(b)

FIG. 3



(a)



(b)

FIG. 4

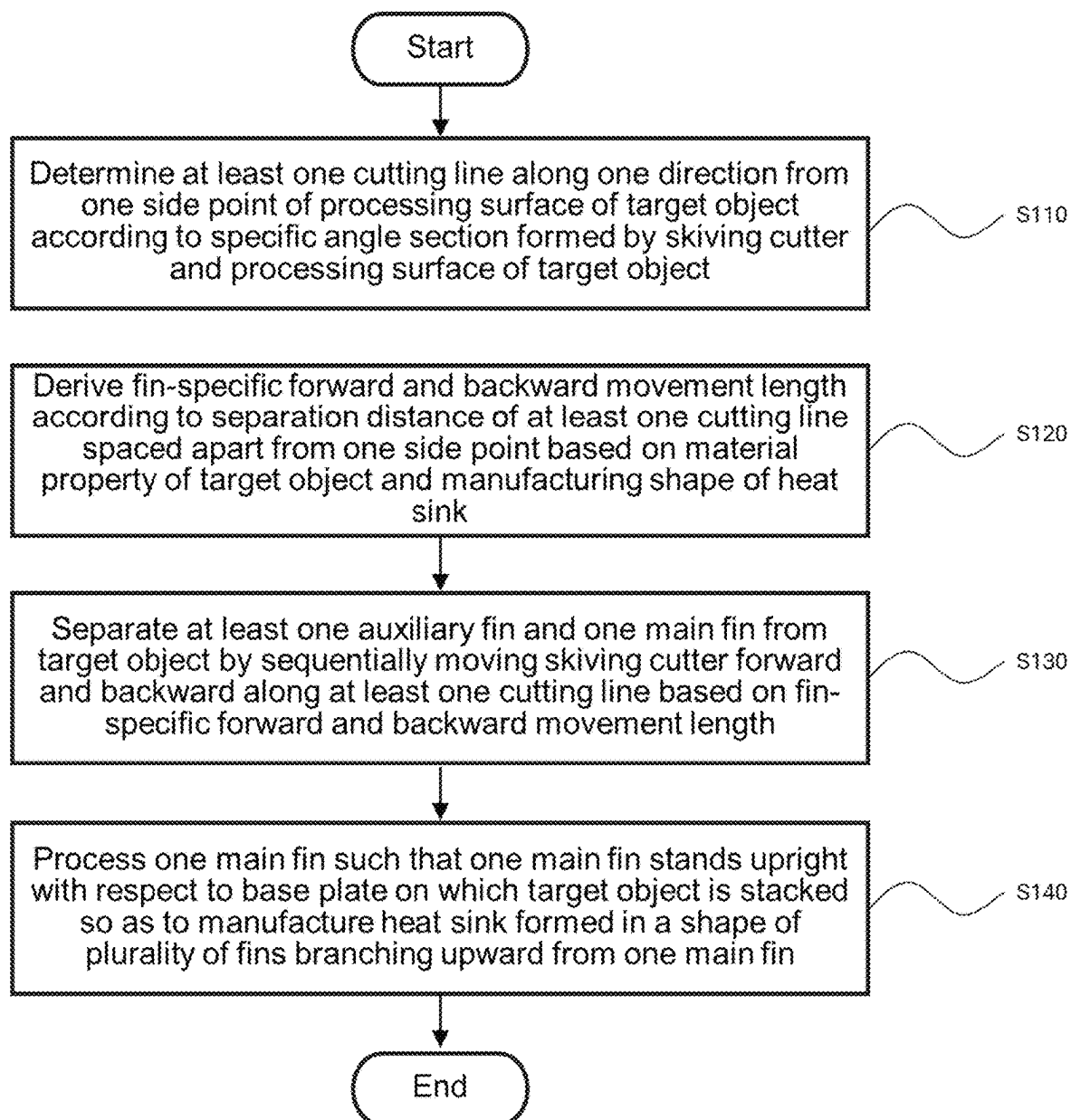


FIG. 5

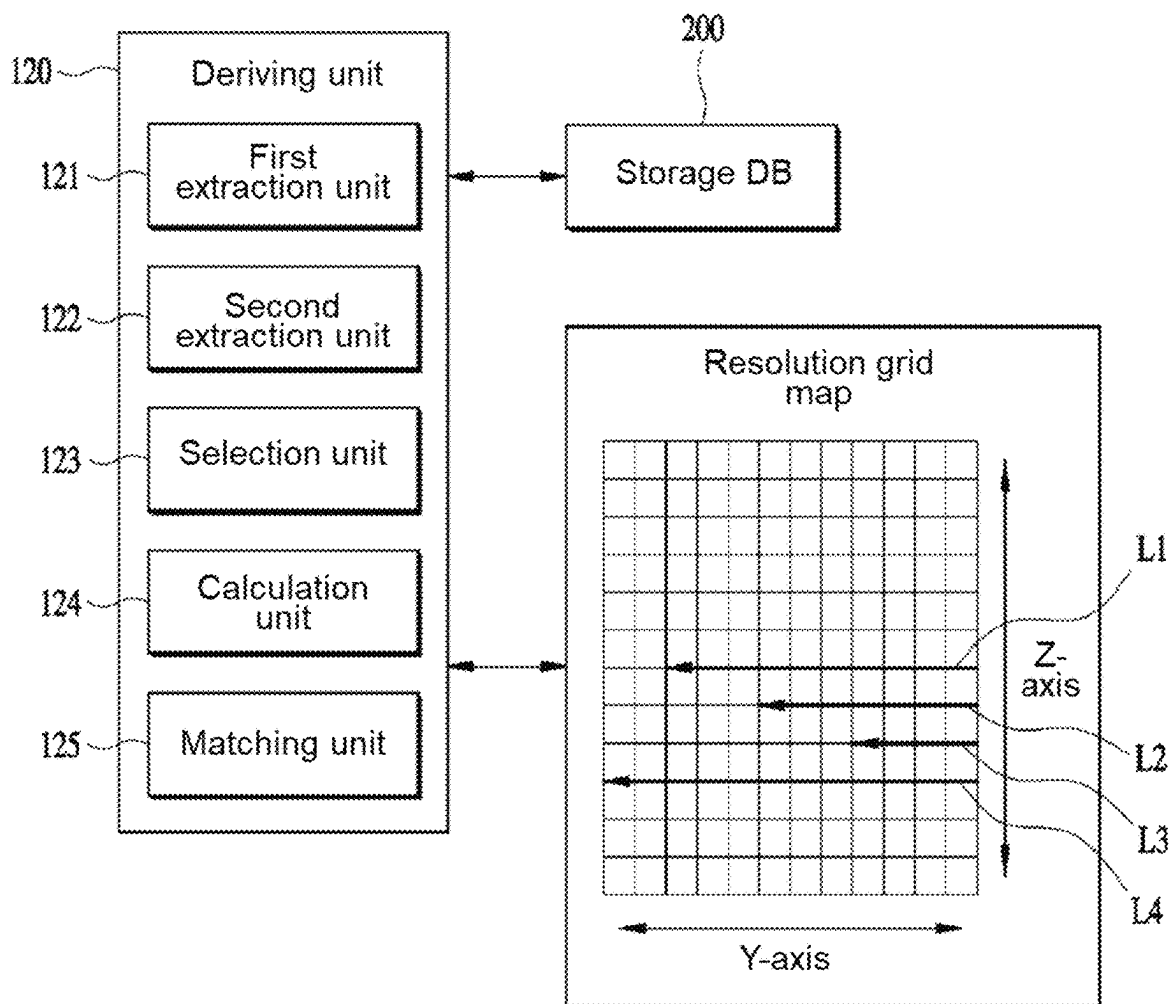


FIG. 6

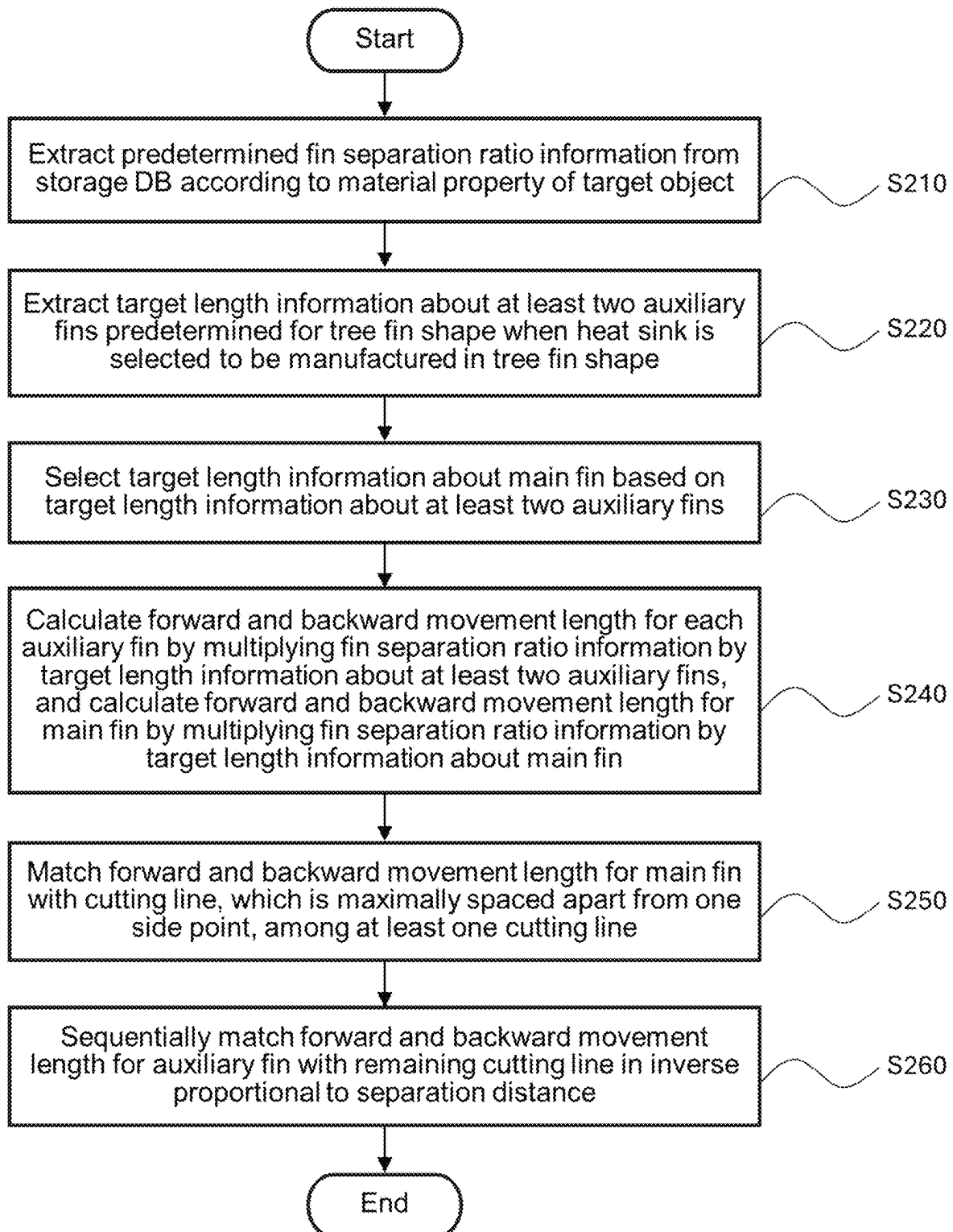
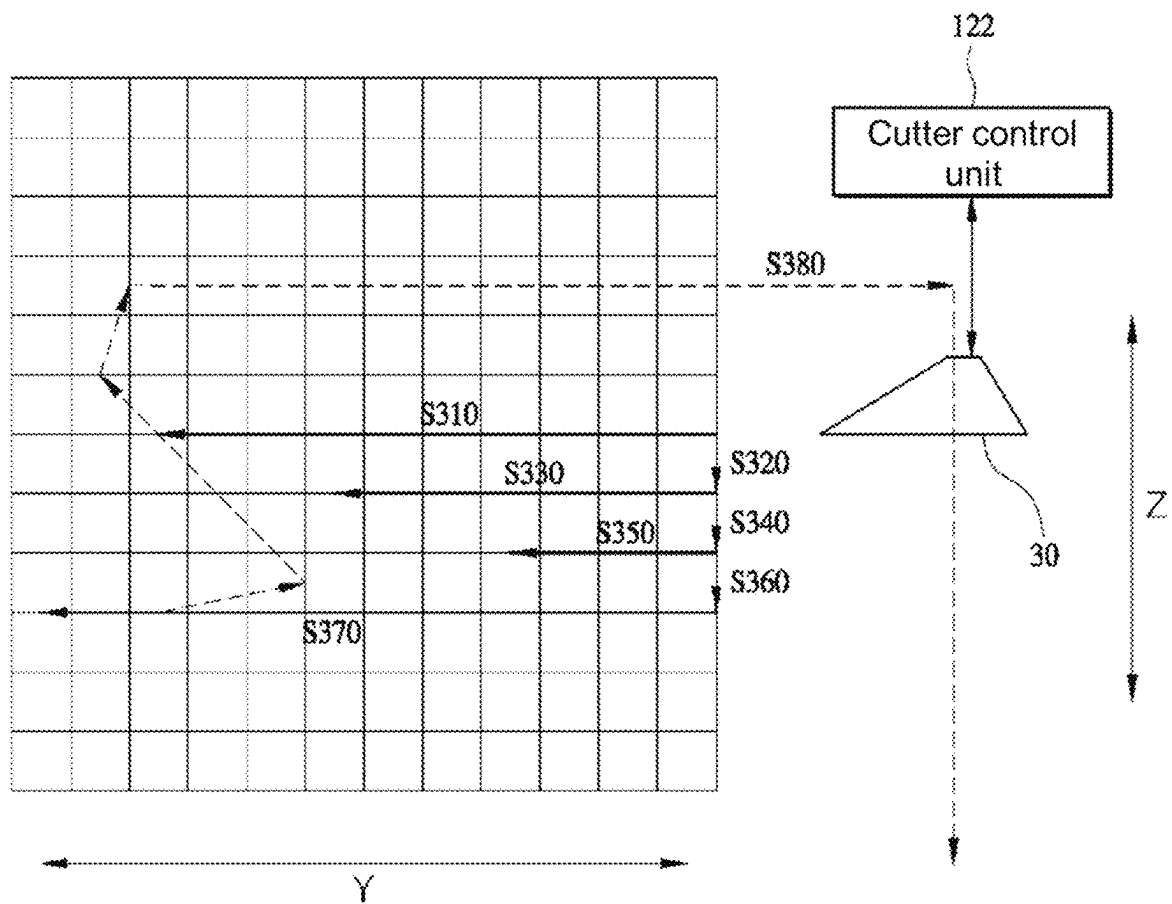


FIG. 7



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HEAT SINK MANUFACTURING METHOD

This application is a National Phase of PCT Patent Application No. PCT/KR2022/020190 having International filing date of Dec. 13, 2022, which claims the benefit of priority of Korean Patent Application No. 10-2021-0177687, filed Dec. 13, 2021, the contents of which are all incorporated herein by reference in their entirety.

TECHNICAL FIELD

The present invention relates to a method and a system for manufacturing a heat sink, and more specifically, to a method and a system for manufacturing a heat sink to provide support such that a heat sink with enhanced heat dissipation efficiency may be rapidly and efficiently manufactured.

BACKGROUND ART

The heat sink is a device (e.g., a heat sink) that effectively discharges heat generated from a heating element using natural convection or forced convection to prevent the heating element from being excessively overheated.

The conventional heat sink is manufactured by attaching an extrusion material or fins to the extrusion material, and in this case, there is no problem if the size of the heat sink is large or the mounting space of the heat sink is large, but in order for the heating element to be small and the heat sink to be compactly designed, it is difficult to expect sufficient heat dissipation characteristics from the extrusion material itself.

That is, due to the technical limitations of the extrusion material, there is a limitation in maximizing a heat dissipation area by reducing the thickness of the fin, improving a tongue ratio, and increasing the number of fins.

To overcome these disadvantages, a skive-type heat sink has been developed. In the case of a skive-type heat sink, it is possible to manufacture the fin much thinner than the extrusion material and to easily adjust a pitch interval.

However, in the case of the conventional skive-type heat sink, it is difficult to obtain an additional heat dissipation area, and thermal interference between the fins disposed at minimum intervals frequently occurs, so that a phenomenon of heat partial stagnation may not be prevented.

DISCLOSURE**Technical Problem**

To solve the above problems, an object of the present invention is to provide a method and a system for manufacturing a heat sink capable of rapidly and efficiently manufacturing the heat sink having excellent heat dissipation area efficiency.

The above and other objects and advantages of the present invention will become apparent from the following description describing preferred embodiments.

Technical Solution

To achieve the above-object, a method for manufacturing a heat sink for processing a target object using a skiving cutter according to one embodiment of the present invention includes: determining, based on information about an angle formed by the skiving cutter and a processing surface of the target object, at least one cutting line along one direction

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from one side point of the processing surface; individually deriving, based on material properties of the target object and a shape of the heat sink to be manufactured, fin-specific forward and backward movement lengths of the skiving cutter according to a separation distance of the at least one cutting line from the one side point; separating, based on the fin-specific forward and backward movement lengths, at least one auxiliary fin and one main fin from the target object by sequentially moving the skiving cutter forward and backward along the at least one cutting line; and processing the one main fin such that the one main fin stands upright with respect to a base plate on which the target object is stacked, so as to manufacture the heat sink formed in a shape of a plurality of fins branching upward from the one main fin.

In the embodiment, the determining may include: confirming whether the target object is disposed on a stacked surface of the base plate in parallel to the stacked surface of the base plate through a camera; and adjusting the base plate such that the base plate is inclined at a specific angle in order to allow the processing surface of the target object to be directed to the skiving cutter.

In the embodiment, the determining may include determining a number of the at least one cutting line based on a number of auxiliary fins identified according to a specific shape of the heat sink to be manufactured.

In the embodiment, the deriving may include: extracting predetermined fin separation ratio information according to the material properties of the target object; extracting target length information about at least two auxiliary fins when the heat sink is selected to have a tree fin shape; selecting target length information about the main fin based on the target length information about the at least two auxiliary fins; calculating a forward and backward movement length for an auxiliary fin by multiplying the fin separation ratio information by the target length information about the at least two auxiliary fins, and calculating a forward and backward movement length of the main fin by multiplying the fin separation ratio information by the target length information about the main fin; and matching the forward and backward movement length for the main fin with a cutting line which is maximally spaced among the at least one cutting line, and matching the forward and backward movement length for the auxiliary fin with a remaining cutting line in inverse proportional to the separation distance.

In the embodiment, the matching may include distinguishably mapping the forward and backward movement length for the main fin and the forward and backward movement length for the auxiliary fin by using a resolution grid map in which the at least one cutting line is set to a Y-axis and intervals between the at least one cutting line spaced apart from the one side point are set to a Z-axis.

Advantageous Effects

According to the embodiment of the present invention, it is possible to rapidly and efficiently manufacture a heat sink having heat dissipation area efficiency that is greater than that in the conventional technology.

DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram schematically showing a heat sink manufacturing system 100 according to one embodiment of the present invention.

FIG. 2(a) is a view showing one embodiment of a heat sink 10, and FIG. 2(b) is a view showing another embodiment of the heat sink 10.

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FIGS. 3(a) and 3(b) are views for explaining operation examples of a skiving cutter 30 of FIG. 1.

FIG. 4 is an operation process of the heat sink manufacturing system 100 of FIG. 1.

FIG. 5 is a block diagram showing a deriving unit 120 of FIG. 1 in more detail, and FIG. 6 is an operation process of the deriving unit 120 of FIG. 5.

FIG. 7 is an operation process of a cutter control unit 120 for processing the heat sink 10 having a tree fin shape.

MODE FOR INVENTION

Hereinafter, the present invention will be described in more detail with reference to the embodiments and drawings of the present invention. The embodiments are only presented exemplarily to describe the present invention in more detail, and it will be apparent to those skilled in the art that the scope of the present invention is not limited by the embodiments.

In addition, unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by those skilled in the art to which the present invention pertains, and in case of conflict, the description herein including the definition will take precedence.

In order to clearly describe the invention proposed in the drawings, parts not related to the description are omitted, and similar parts are denoted by similar reference numerals throughout the specification. In addition, when one component “includes” some elements, it does not exclude other elements, but may further include the other elements, unless particularly stated otherwise. In addition, the term “unit” described in the specification means one unit or block that performs a specific function.

In each step, the reference numerals (first, second, etc.) are used for convenience of description, and the reference numerals do not describe the order of each step, and each step may be performed differently from the specified order unless the specific order is clearly described in the context. That is, the respective steps may be performed in the same order as described above, may be performed substantially simultaneously, or may be performed in the reverse order.

FIG. 1 is a block diagram schematically showing a heat sink manufacturing system 100 according to one embodiment of the present invention, FIG. 2(a) is a view showing one embodiment of a heat sink 10, FIG. 2(b) is a view showing another embodiment of the heat sink 10, and FIGS. 3(a) and 3(b) are views for explaining operation examples of a skiving cutter 30 of FIG. 1.

Referring to FIGS. 1 to 3(b), the heat sink manufacturing system 100 according to one embodiment of the present invention may include a determination unit 110, a deriving unit 120, and a cutter control unit 130 in order to process a target object 1 by using the skiving cutter 30.

First, the determination unit 110 may determine at least one cutting line (e.g., L1 to L4) positioned along one direction from one side point (e.g., P1) of a processing surface of the target object 1 according to a specific angle section (e.g., θ) formed by the skiving cutter 30 and the processing surface of the target object 1.

At least one cutting line (e.g., L1 to L4) used herein is a vertical line spaced apart from one side point (e.g., P1) of the processing surface at the same interval along one direction (e.g., Z-axis direction), and may be a guide line for moving the skiving cutter 30 forward and backward. In this case, at

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least one cutting line (e.g., L1 to L4) may be determined to be positioned in parallel to each other along the Z-axis direction.

Further, the determination unit 110 may determine the number (e.g., $n+1$) of at least one cutting line (e.g., L1 to L4) based on the number (n) of auxiliary fins identified according to a specific shape of the heat sink 10 to be manufactured.

For example, when the number of auxiliary fins identified according to the specific shape of the heat sink 10 to be manufactured is 1, the determination unit 110 may determine the number of at least one cutting line (e.g., L1 to L4) as 2. Further, when the number of auxiliary fins identified according to the specific shape of the heat sink 10 to be manufactured is 3, the determination unit 110 may determine the number of at least one cutting line (e.g., L1 to L4) as 4.

In this case, as shown in FIGS. 2(a) and 2(b), the heat sink 10 may be manufactured in any one specific shape of any one of a Y fin or a tree fin.

For example, when the specific shape of the heat sink 10 to be manufactured is a Y fin, the determination unit 110 may determine the number of auxiliary fins as 1, and when the specific shape of the heat sink 10 to be manufactured is a tree fin, the determination unit 110 may determine the number of auxiliary fins as 3.

The heat sink 10 manufactured in any one specific shape of the Y fin or the tree fin may be coupled to a component or the like that generates heat inside an electric device to efficiently dissipate the heat generated in an element due to air passing between heat dissipation fins or the flow of a fluid for cooling.

That is, as the target object 1 is stacked on a base plate 20, the determination unit 110 may adjust the base plate 20 at a predetermined angle such that the processing surface of the target object 1 is obliquely directed to the skiving cutter 30, and then may determine at least one cutting line (e.g., L1 to L4) positioned along one direction from one side point (e.g., P1) of the processing surface of the target object 1.

The determination unit 110 according to the embodiment may confirm whether the target object 1 is disposed on a stacked surface of the base plate 20 in parallel to the stacked surface of the base plate 20 through a camera (not shown). In this case, the determination unit 110 may adjust the base plate 20 such that the base plate 20 is inclined at a specific angle in order to allow the processing surface of the target object 1 is directed to the skiving cutter 30.

Next, the deriving unit 120 may individually derive a fin-specific forward and backward movement length of the skiving cutter 30, which correspond to a separation distance of at least one cutting line (e.g., L1 to L4) from one side point (e.g., P1), based on manufacturing characteristics of the target object 1 and a manufacturing shape of the heat sink 10.

The fin-specific forward and backward movement length used herein may mean a cutting length of the target object 1 as the skiving cutter 30 moves forward and backward.

For example, when the material property of the target object 1 is copper and the shape of the heat sink 10 is a tree fin, the deriving unit 120 may derive the forward and backward movement length of the skiving cutter 30, which corresponds to a first cutting line spaced apart from one side point (e.g., P1) by 1.5 mm, as 10 mm, may derive the forward and backward movement length of the skiving cutter 30, which corresponds to a second cutting line spaced apart from one side point (e.g., P1) by 3.0 mm, as 7 mm, may derive the forward and backward movement length of the

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skiving cutter **30**, which corresponds to a third cutting line spaced apart from one side point (e.g., **P1**) by 4.5 mm, as 4 mm, and may derive the forward and backward movement length of the skiving cutter **30**, which corresponds to a fourth cutting line spaced apart from one side point (e.g., **P1**) by 6.0 mm, as 12 mm. Hereinafter, the deriving unit **120** will be described in more detail with reference to FIGS. **4** to **6**.

Next, the cutter control unit **130** may sequentially move the skiving cutter **30** forward and backward along at least one cutting line (e.g., **L1** to **L4**) based on the fin-specific forward and backward movement length to separate at least one auxiliary fin **12_1** to **12_N** and one main fin **11** from the target object **1**.

That is, as shown in FIGS. **3a** and **3b**, the cutter control unit **130** may form at least one auxiliary fin (**12_1** to **12_N**) and one main fin **11** with respect to the target object **1** by using the skiving cutter **30** that moves according to the fin-specific forward and backward movement length.

Thereafter, the cutter control unit **130** may process one main pin **11** using the skiving cutter **30** such that one main fin **11** stands upright with respect to the base plate **20** on which the target object **1** is stacked, so as to manufacture the heat sink **10** formed in a shape of a plurality of fins branching upward from one main fin **11**.

That is, the cutter control unit **130** may allow one main fin **11** to stand upright, thereby manufacturing the heat sink **10** having a tree fin or Y fin shape.

The heat sink manufacturing system **100** according to the embodiment of the present invention may be sequentially operated according to a predetermined heat sink manufacturing order through the determination unit **110**, the deriving unit **120**, and the cutter control unit **130** to support rapid and effective manufacturing of the heat sink **10** having higher heat dissipation efficiency.

Hereinafter, the configuration of the present invention and the effects thereof will be described in more detail with reference to specific examples and comparative examples. However, the present embodiment is intended to describe the present invention in more detail, and the scope of the present invention is not limited to these embodiments.

FIG. **4** is an operation process of the heat sink manufacturing system **100** of FIG. **1**.

Referring to FIGS. **1** to **4**, in step **S110**, first, the determination unit **110** may determine at least one cutting line (e.g., **L1** to **L4**) along one direction from one side point (e.g., **P1**) of the processing surface of the target object **1** according to the specific angle section (e.g., **0**) formed by the skiving cutter **30** and the processing surface of the target object **1**.

According to one embodiment, the determination unit **110** may confirm whether the target object **1** is disposed on the stacked surface of the base plate **20** in parallel to the stacked surface of the base plate **20** through a camera (not shown) before step **S110**. In this case, the determination unit **110** may adjust the base plate **20** such that the base plate **20** is inclined at a specific angle in order to allow the processing surface of the target object **1** is directed to the skiving cutter **30**.

According to another embodiment, the determination unit **110** may determine the number (e.g., $n+1$) of at least one cutting line (e.g., **L1** to **L4**) based on the number (n) of auxiliary fins identified according to a specific shape of the heat sink **10** to be manufactured.

For example, when the heat sink **10** is selected to be manufactured in a tree fin shape, the determination unit **110** may determine the number ($n+1$) of at least one cutting line (e.g., **L1** to **L4**) as 4 based on the number of auxiliary fins of 3, which is identified according to the tree fin shape.

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Thereafter, in step **S120**, the deriving unit **120** may individually derive a fin-specific forward and backward movement length of the skiving cutter **30** according to a separation distance of at least one cutting line (e.g., **L1** to **L4**) from one side point (e.g., **P1**), based on manufacturing characteristics of the target object **1** and a manufacturing shape of the heat sink **10**.

Next, in step **S130**, the cutter control unit **130** may sequentially move the skiving cutter **30** forward and backward along at least one cutting line (e.g., **L1** to **L4**) based on the fin-specific forward and backward movement length to separate at least one auxiliary fin **12_1** to **12_N** and one main fin **11** from the target object **1**.

Thereafter, in step **S140**, the cutter control unit **130** may process one main pin **11** using the skiving cutter **130** such that one main fin **11** stands upright with respect to the base plate **20** on which the target object **1** is stacked, so as to manufacture the heat sink **10** formed in a shape of a plurality of fins branching upward from one main fin **11**, for example, in a tree fin or Y fin shape.

FIG. **5** is a block diagram showing a deriving unit **120** of FIG. **1** in more detail, and FIG. **6** is an operation process of the deriving unit **120** of FIG. **5**.

Referring to FIGS. **5** and **6**, the deriving unit **120** may include first and second extraction units **121** and **122**, a selection unit **123**, a calculation unit **124**, and a matching unit **125**.

First, in step **S210**, the first extraction unit **121** may extract fin separation ratio information from a storage DB **200** according to the material property of the target object **1**.

The fin separation ratio information used herein is values obtained by digitizing a ratio of an actual separation area to a total cutting area of the target object **1** to be cut by the skiving cutter **30** according to the material property through an experiment, and may be classified and managed for each material in the storage DB **200**.

For example, the fin separation ratio information of the material property corresponding to copper may correspond to 1.8, and the fin separation ratio information of the material property corresponding to aluminum may correspond to 2.0.

Next, in step **S220**, when the heat sink **10** is selected to be manufactured in a tree fin shape, the second extraction unit **122** may extract target length information about at least two auxiliary fins predetermined for a tree fin shape from the storage DB **200**. For example, when the heat sink **10** is selected to be manufactured in a tree fin shape including the first to third auxiliary fins **12_1** to **12_3**, the second extraction unit **122** may extract predetermined target length information about at least two auxiliary fins from the storage DB **200** as 5.5 mm, 3.9 mm, and 2.2 mm.

Next, in step **S230**, the selection unit **123** may select target length information about the main fin based on the target length information about at least two auxiliary fins.

In this case, the target length information about the main fin may be selected to be greater than the target length information about at least two auxiliary fins. For example, when the target length information about at least two auxiliary fins is 5.5 mm, 3.9 mm, and 2.2 mm, the selection unit **123** may select the target length information about the main pin as a 6.6 mm that is greater than the target length information about at least two auxiliary pins.

Thereafter, in step **S240**, the calculation unit **124** may calculate a forward and backward movement length for each auxiliary fin by multiplying the predetermined fin separation ratio information by the target length information about at least two auxiliary fins. In addition, the calculation unit **124** may simultaneously calculate the forward and backward

movement length for the main fin by multiplying the fin separation ratio information by the target length information about the main fin.

Specifically, when the specific shape of the heat sink **10** to be manufactured corresponds to a tree fin shape, the calculation unit **124** may calculate a first forward and backward movement length of the skiving cutter **30** with respect to the first auxiliary pin **12_1** by multiplying target length information about the first auxiliary fin **12_1** among at least one auxiliary fins **12_1** to **12_N** by the fin separation ratio information. For example, when the fin separation ratio information is 1.8 and the target length information about the first auxiliary fin **12_1** is 5.5 mm, the calculation unit **124** may calculate the first forward and backward movement length as 10 mm.

Moreover, the calculation unit **124** may calculate a second forward and backward movement length of the skiving cutter **30** with respect to the second auxiliary pin **12_2** by multiplying target length information about the second auxiliary fin **12_2** among at least one auxiliary fins **12_1** to **12_N** by the fin separation ratio information. For example, when the fin separation ratio information is 1.8 and the target length information about the second auxiliary fin **12_2** is 3.9 mm, the calculation unit **124** may calculate the second forward and backward movement length as 7 mm.

Moreover, the calculation unit **124** may calculate a third forward and backward movement length of the skiving cutter **30** with respect to the third auxiliary pin **12_3** by multiplying target length information about the third auxiliary fin **12_3** among at least one auxiliary fins **12_1** to **12_N** by the fin separation ratio information. For example, when the fin separation ratio information is 1.8 and the target length information about the third auxiliary fin **12_3** is 2.2 mm, the calculation unit **124** may calculate the third forward and backward movement length as 4 mm.

In this case, the calculation unit **124** may calculate a fourth forward and backward movement length of the skiving cutter **30** with respect to the main fin **11** by multiplying the target length information about the main fin **11** by the fin separation ratio information. For example, when the fin separation ratio information is 1.8 and the target length information about the main fin **11** is 6.6 mm, the calculation unit **124** may calculate the fourth forward and backward movement length as 12 mm.

According to the embodiment, when the specific shape of the heat sink **10** to be manufactured corresponds to a Y fin shape, the calculation unit **124** may skip the step of calculating the second and third forward and backward movement lengths from among the first to fourth forward and backward movement lengths.

Thereafter, in **S250**, the matching unit **125** may match the forward and backward movement length of the main fin with the cutting line (e.g., **L4**), which is maximally spaced apart from one side point (e.g., **P1**), among at least cutting line (e.g., **L1** to **L4**) through the calculation unit **124**. For example, the matching unit **125** may match the forward and backward movement length (12 mm) for the main fin with a maximum separation distance (6.0 mm) of at least one cutting line (e.g., **L1** to **L4**).

Thereafter, the matching unit **125** may sequentially match the forward and backward movement length for the auxiliary fin with a remaining cutting line (e.g., **L1** to **L3**) among at least one cutting line (e.g., **L1** to **L4**) in inverse proportional to the separation distance. For example, the matching unit **125** may match the forward and backward movement length (55 mm: 3.9 mm: 2.2 mm) for the auxiliary fin with the

remaining cutting line (e.g., **L1** to **L3**) in inverse proportional to a remaining separation distance (1.5 mm: 3.0 mm: 4.5 mm).

According to the embodiment, the matching unit **125** may manage the forward and backward movement length of the main fin and the forward and backward movement length of the auxiliary fin by distinguishing mapping the same with a grid length by using a resolution grid map in which at least one cutting line (e.g., **L1** to **L4**) is set to a Y-axis and intervals between at least one cutting line (e.g., **L1** to **L4**) separated from one side point (e.g., **P1**) are set to a Z-axis.

FIG. 7 is an operation process of the cutter control unit **120** for processing the heat sink **10** having a tree fin shape.

Referring to FIGS. 1, 2(a), and 7, first, in step **S310**, the cutter control unit **130** may allow the skiving cutter to move forward and backward in a Y-axis direction by the first forward and backward movement length (e.g., **Y1** to **Y10.5**), which is derived through the deriving unit **120**, from grid coordinates (e.g., **Z7**, **Y1**) of the first cutting line confirmed through the resolution grid map. In this case, the first auxiliary fin (e.g., **12_1** of FIG. 2(a)) may be separated from the target object **1**.

Next, in step **S320**, the cutter control unit **130** may move the skiving cutter in a Z-axis direction from grid coordinates (e.g., **Z8**, **Y1**) of the second cutting line confirmed through the resolution grid map.

Next, in step **S330**, the cutter control unit **130** may allow the skiving cutter to move forward and backward in the Y-axis direction by the second forward and backward movement length (e.g., **Y1** to **Y7.5**), which is derived through the deriving unit **120**, from grid coordinates (e.g., **Z8**, **Y1**) of the second cutting line confirmed through the resolution grid map. In this case, the second auxiliary fin (e.g., **12_2** of FIG. 2(a)) may be separated from the target object **1**.

Next, in step **S340**, the cutter control unit **130** may move the skiving cutter in the Z-axis direction from grid coordinates (e.g., **Z9**, **Y1**) of the third cutting line confirmed through the resolution grid map.

Next, in step **S350**, the cutter control unit **130** may allow the skiving cutter to move forward and backward in the Y-axis direction by the third forward and backward movement length (e.g., **Y1** to **Y4.5**), which is derived through the deriving unit **120**, from grid coordinates (e.g., **Z9**, **Y1**) of the third cutting line confirmed through the resolution grid map. In this case, the third auxiliary fin (e.g., **12_3** of FIG. 2(a)) may be separated from the target object **1**.

Next, in step **S360**, the cutter control unit **130** may move the skiving cutter in the Z-axis direction from grid coordinates (e.g., **Z10**, **Y1**) of the fourth cutting line confirmed through the resolution grid map.

Next, in step **S370**, the cutter control unit **130** may allow the skiving cutter to move forward and backward in the Y-axis direction by the fourth forward and backward movement length (e.g., **Y1** to **Y12.5**), which is derived through the deriving unit **120**, from grid coordinates (e.g., **Z10**, **Y1**) of the fourth cutting line confirmed through the resolution grid map. In this case, the fourth auxiliary fin (e.g., **11** of FIG. 2(a)) may be separated from the target object **1**.

Thereafter, in step **S380**, the cutter control unit **130** may process the main fin (for example, **11** of FIG. 2(a)) such that the main fin stands upright, thereby manufacturing the heat sink **10** of a tree fin shape.

In the present specification, only few examples of various embodiments performed by the present inventors have been described, but the technical idea of the present invention is not limited thereto, and may be modified by those skilled in the art to be variously performed.

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The invention claimed is:

1. A method for manufacturing a heat sink for processing
a target object using a skiving cutter, the method comprising:
determining, based on information about an angle formed
by the skiving cutter and a processing surface of the
target object, at least one cutting line along one direction
from one side point of the processing surface;
individually deriving, based on material properties of the
target object and a shape of the heat sink to be manu-
factured, fin-specific forward and backward movement
lengths of the skiving cutter according to a separation
distance of the at least one cutting line from the one
side point; and
separating, based on the fin-specific forward and back-
ward movement lengths, at least one auxiliary fin and
at least one main fin from the target object by sequen-
tially moving the skiving cutter forward and backward
along the at least one cutting line,
wherein the at least one auxiliary fin is formed by pro-
cessing the at least one main fin such that the at least
one main fin stands upright with respect to a base
plate on which the target object is stacked, so as to
manufacture the heat sink formed with the at least one
main fin and the at least one auxiliary fin branching
upward from the at least one main fin,
wherein the deriving includes:
extracting predetermined fin separation ratio information
according to the material properties of the target object;
extracting target length information about at least two
auxiliary fins such that the heat sink has a tree fin shape;
selecting target length information about the at least one
main fin based on the target length information about
the at least two auxiliary fins;
calculating a forward and backward movement length for
an auxiliary fin by multiplying the fin separation-ratio

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information by the target length information about the
at least two auxiliary fins, and calculating a forward and
backward movement length of the at least one main fin
by multiplying the fin separation ratio information by
the target length information about the at least one main
fin; and

matching the forward and backward movement length for
the at least one main fin with a cutting line which is
maximally spaced apart from one side point (P1)
among the at least one cutting line, and matching the
forward and backward movement length for the auxil-
iary fin with a remaining cutting line in inverse pro-
portional to the separation distance.

2. The method of claim 1, wherein the determining
includes:

confirming whether the target object is disposed on a
stacked surface of the base plate in parallel to the
stacked surface of the base plate through a camera; and
adjusting the base plate such that the base plate is inclined
at a specific angle in order to allow the processing
surface of the target object to be directed to the skiving
cutter.

3. The method of claim 1, wherein the determining
includes determining a number of the at least one cutting line
based on a number of auxiliary fins identified according to
a specific shape of the heat sink to be manufactured.

4. The method of claim 1, wherein the matching includes
distinguishably mapping the forward and backward move-
ment length for the at least one main fin and the forward and
backward movement length for the at least two auxiliary fins
auxiliary fin by using a resolution grid map in which the at
least one cutting line is set to a Y-axis and intervals between
the at least one cutting line spaced apart from the one side
point are set to a Z-axis.

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